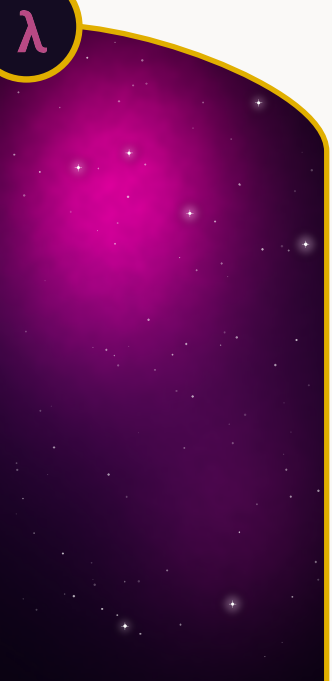


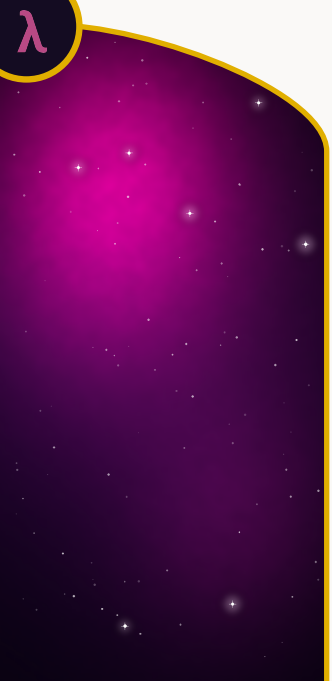
L11A: Streams II

CS1101S: Programming Methodology

Martin Henz

28th October, 2026

Version 1.4 — Built: 2026-10-28 09:18:57 SGT



Announcements

- Mastery Check 2 begins now
- Topics
  - environment model
  - memoization
  - streams

Announcements

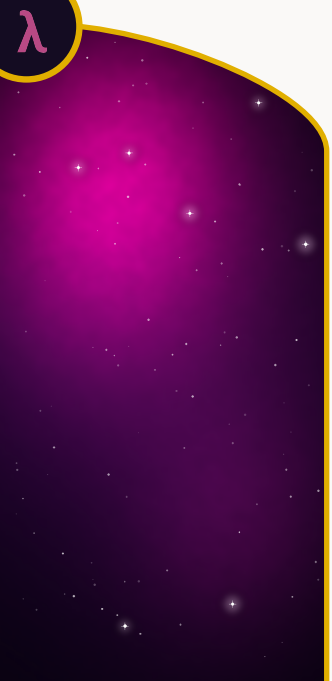
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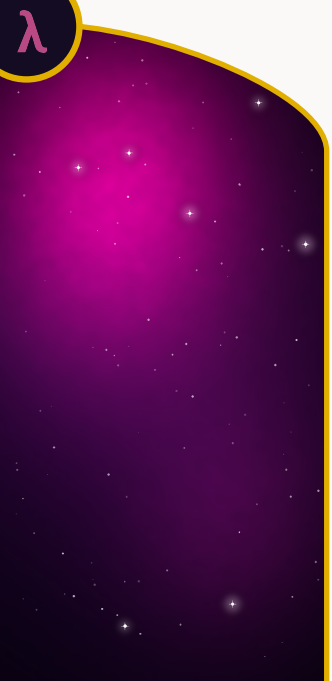
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- Forum hours: one last session next Monday, 6:30pm, COM1 basement

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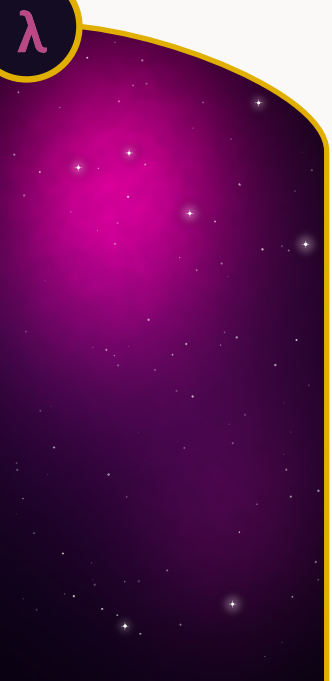


Course Overview



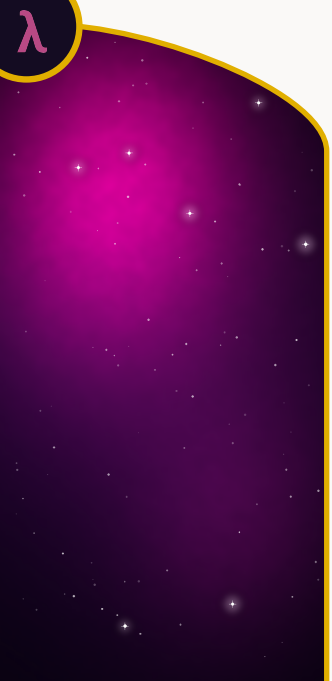
Course Overview

- Unit 1—Functions (textbook Chapter 1)



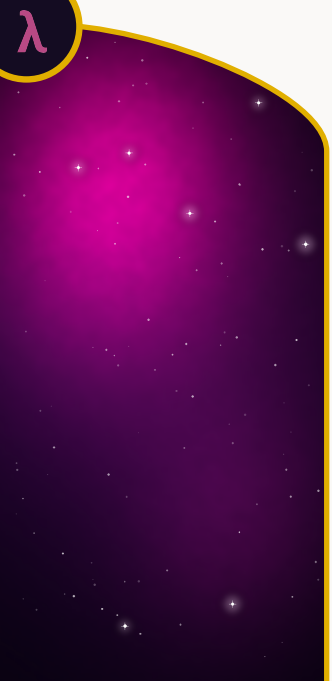
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  - Getting acquainted with the elements of programming, using **functional abstraction**



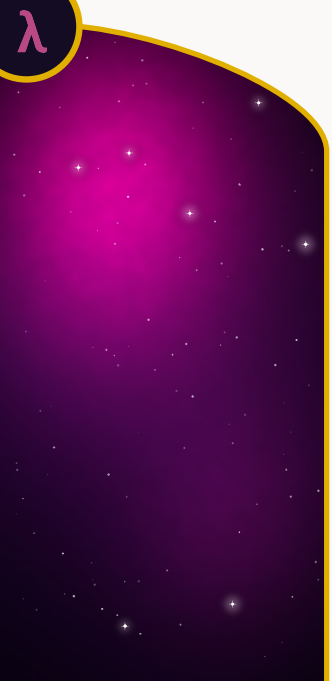
Course Overview

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  - Getting acquainted with the elements of programming, using **functional abstraction**
  - Learning to read programs, and using the **substitution model**



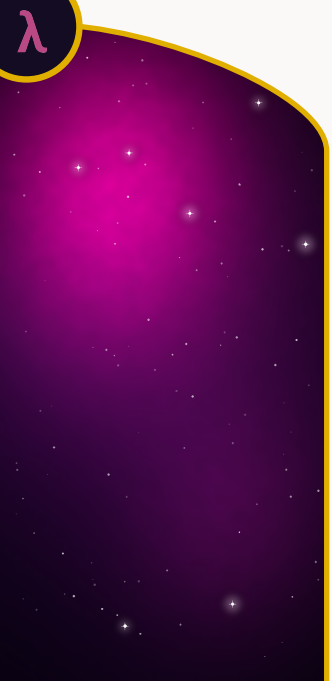
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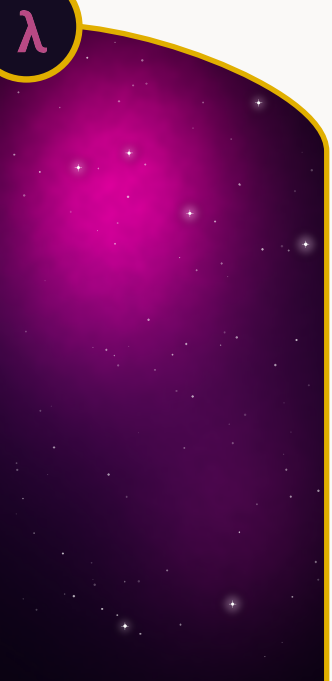
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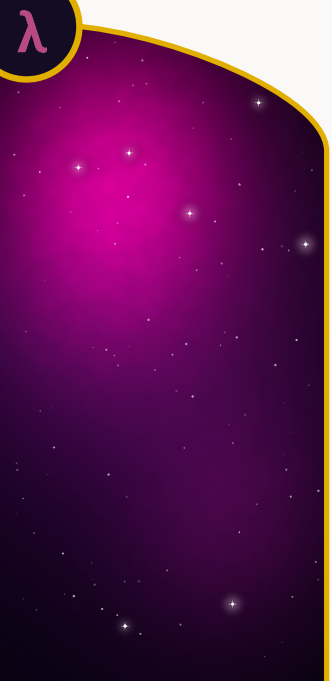
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  - Getting familiar with data: pairs, linked lists, trees, streams



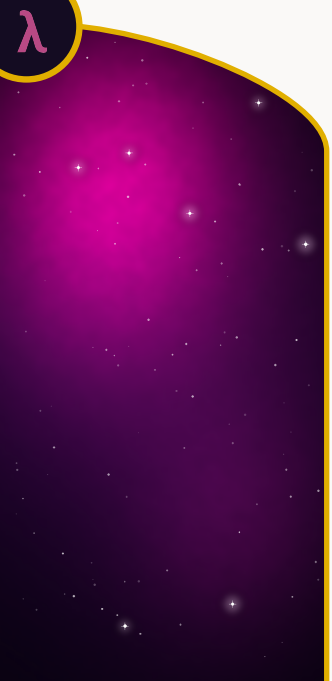
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  - Searching in linked lists and trees, sorting of linked lists

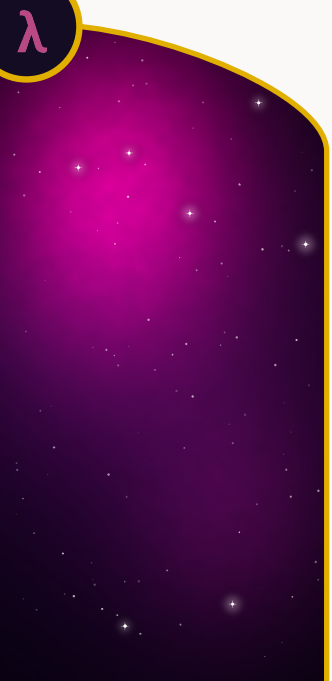


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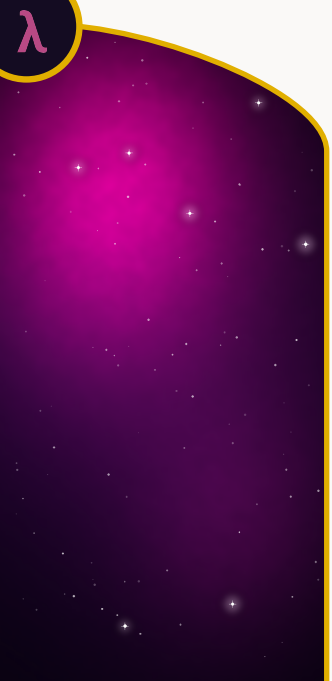


Course Overview, continued



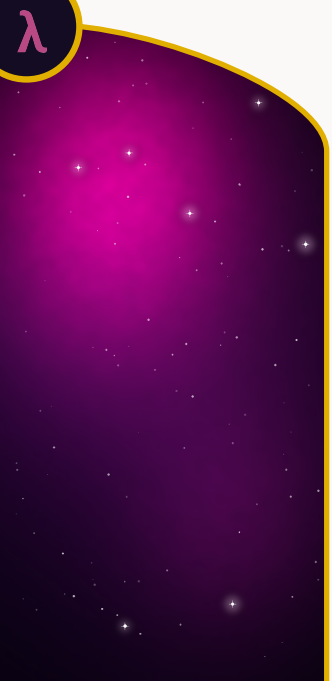
Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)



Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions



Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions
  - Lists, loops, searching in and sorting of lists

Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions
  - Lists, loops, searching in and sorting of lists
  - Reading programs using the environment model

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- Unit 3—State (parts of textbook Chapter 3)
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Course Overview, continued

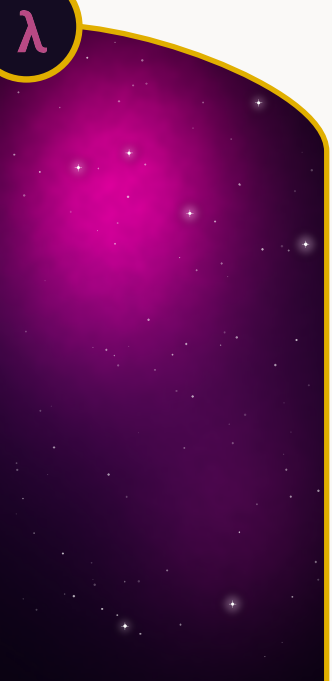
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  - Memoization, **Memoized Streams**

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  - Memoization, **Memoized Streams**
  - Metalinguistic abstraction



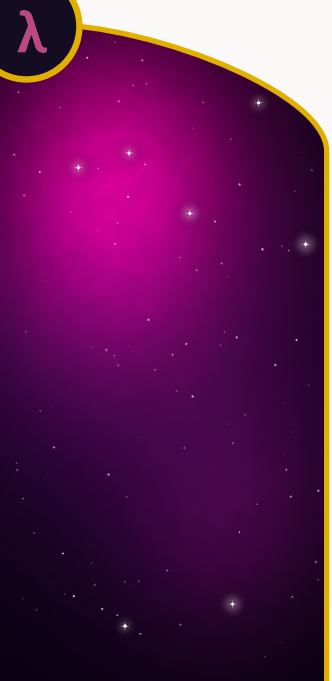
Outline

Review

Streams as Recursive Definitions

Memoization

Applications



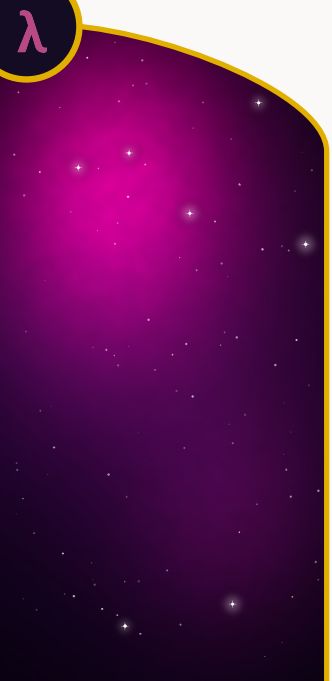
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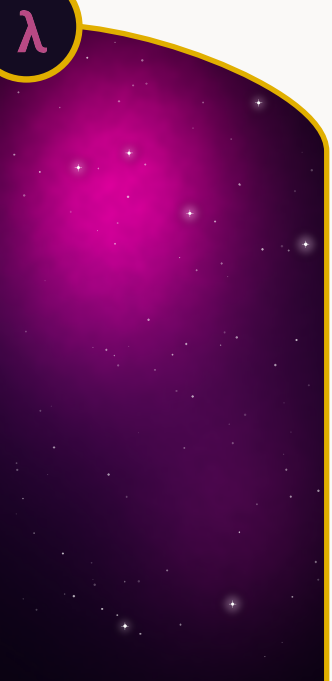
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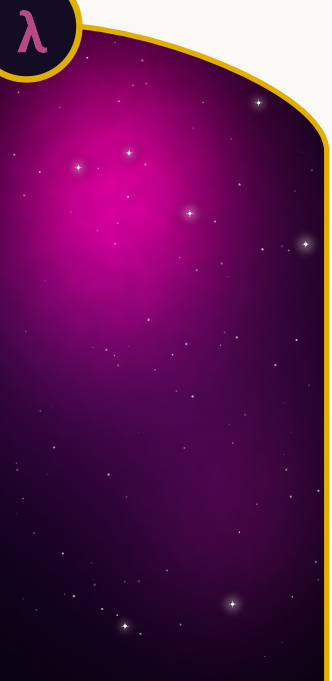


Idea of Streams



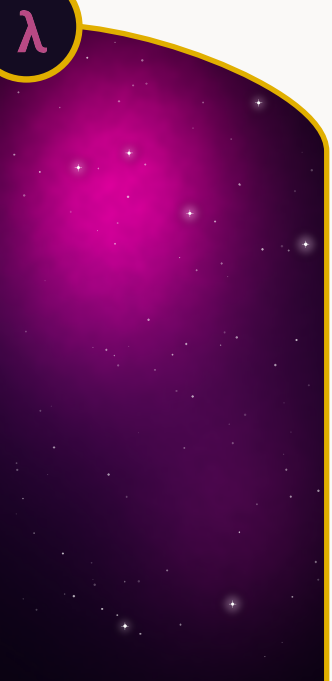
Idea of Streams

- Delayed linked lists



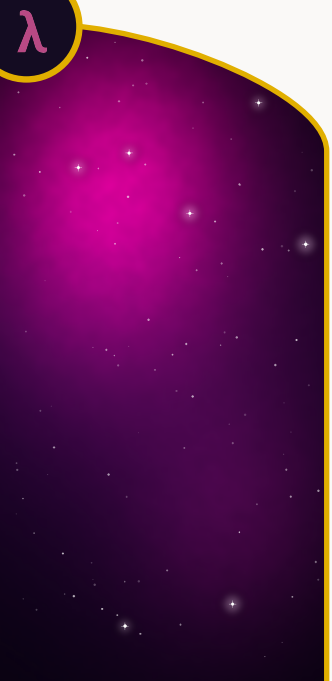
Idea of Streams

- Delayed linked lists
  - Our pairs contain a data item as `head` (as usual), but a function as `tail` that can be activated when needed



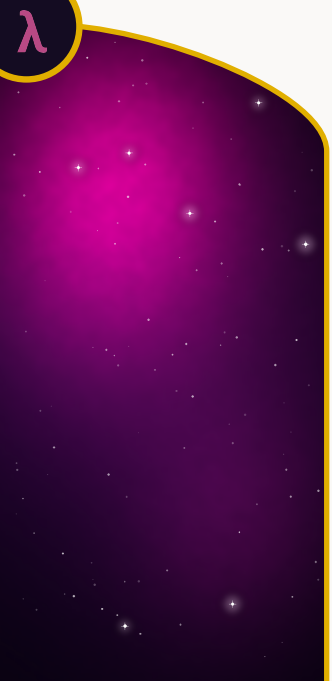
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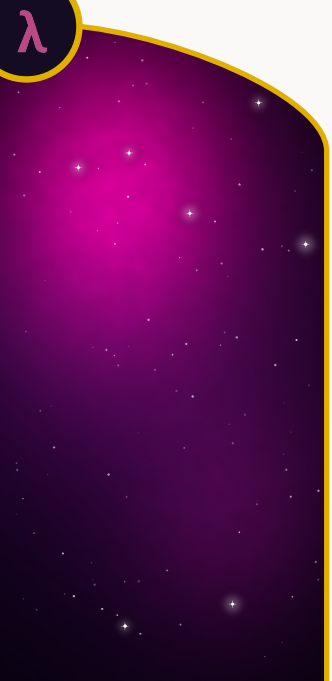
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  - A stream of a certain type is either `None`,



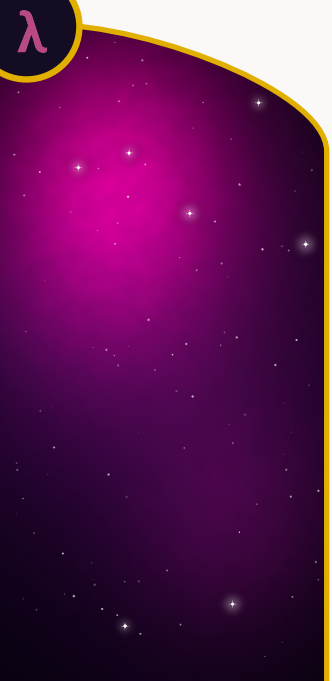
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  - A stream of a certain type is either `None`,
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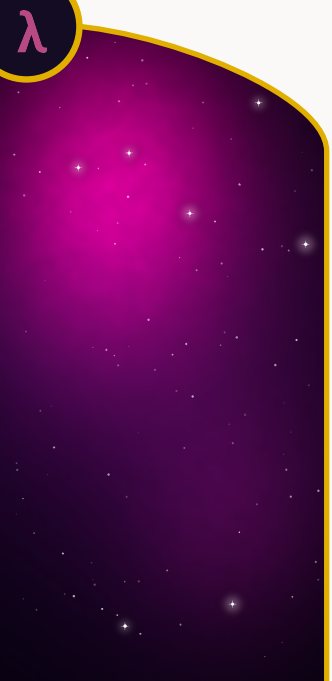
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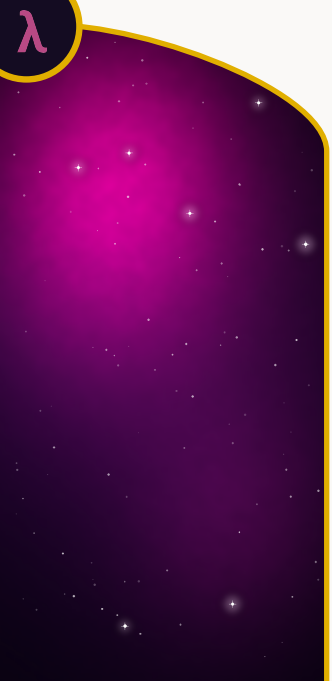
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    - whose `tail` is a nullary function that returns a stream of that type



Idea of Streams

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- Stream discipline



Idea of Streams

- Delayed linked lists
  - Our pairs contain a data item as `head` (as usual), but a function as `tail` that can be activated when needed
- Streams
  - A stream of a certain type is either `None`,
  - or a pair
    - whose `head` is of that type and
    - whose `tail` is a nullary function that returns a stream of that type
- Stream discipline
  - Like the linked-list discipline, now using streams

Examples — Finite Streams

```
# An empty stream
s1 = None
```



Examples — Finite Streams

```
# An empty stream
s1 = None
```



```
# A stream with element 1
s2 = pair(1, lambda: None)
```



Examples — Finite Streams

```
# An empty stream
s1 = None
```

```
# A stream with element 1
s2 = pair(1, lambda: None)
```

```
# A stream with elements 1, 2, 3
s3 = pair(1,
          lambda: pair(2,
                       lambda: pair(3,
                                    lambda: None)))
```

Example — An Infinite Stream

```
def ones_stream():
    return pair(1, ones_stream)
ones = ones_stream()
def stream_tail(stream):
    return tail(stream)()
```



Example — An Infinite Stream

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- head(ones) $\Rightarrow$ 1

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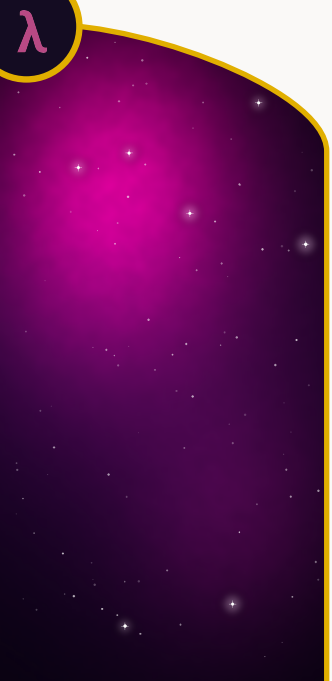
- head(ones) $\Rightarrow$ 1
- head(stream_tail(ones)) $\Rightarrow$ 1

Example — An Infinite Stream

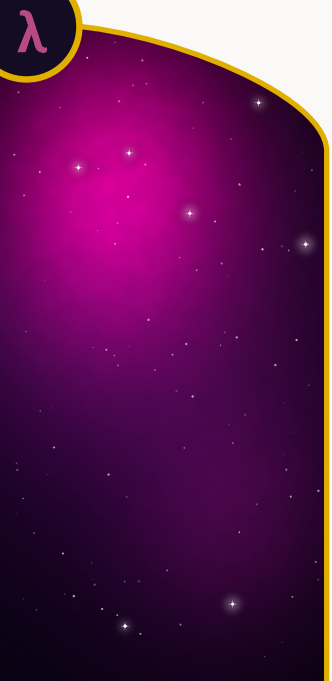
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- head(ones) $\Rightarrow$ 1
- head(stream_tail(ones)) $\Rightarrow$ 1
- head(stream_tail(stream_tail(ones))) $\Rightarrow$ 1

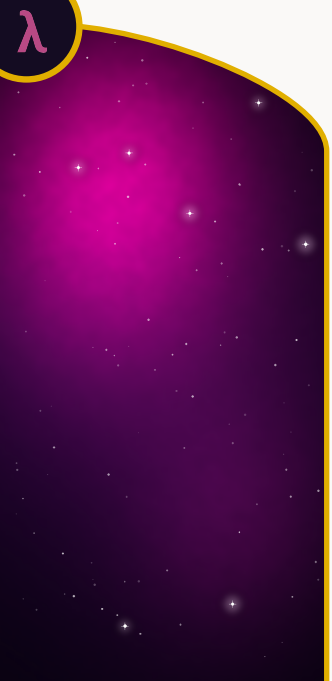


Stream Processing



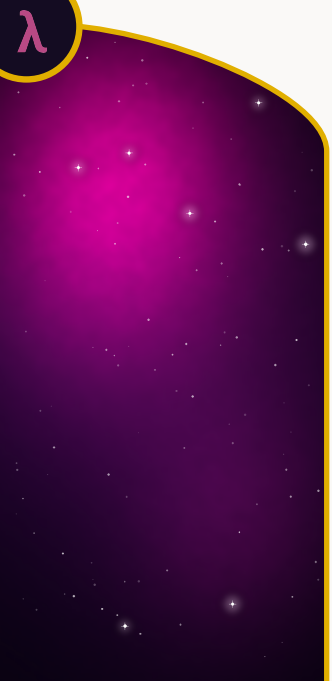
Stream Processing

- Like linked lists, except



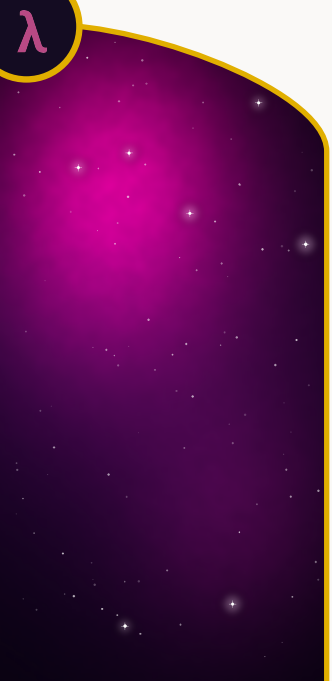
Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function



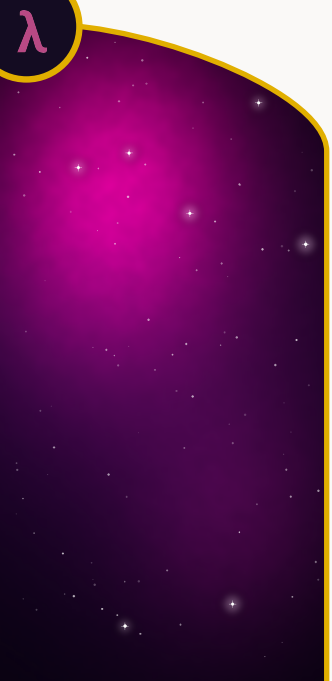
Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`



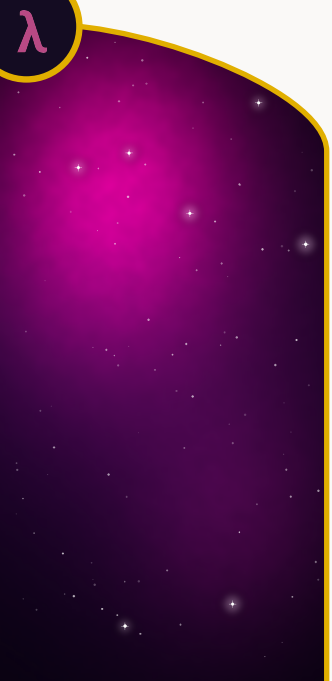
Stream Processing

- Like linked lists, except
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  - Use `stream_tail` instead of `tail`
- Stream support in Python §3



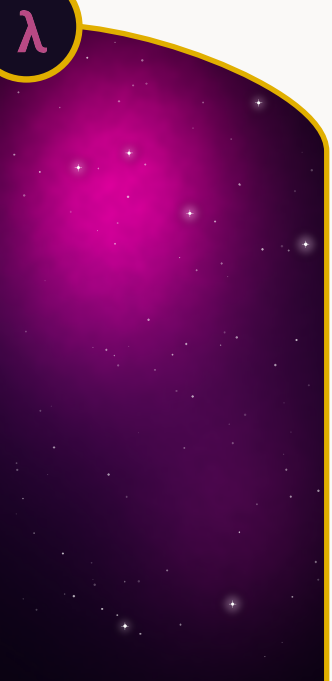
Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`
- Stream support in Python §3
  - Pre-declared functions for stream processing



Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`
- Stream support in Python §3
  - Pre-declared functions for stream processing
    - Examples: `stream_tail`, `stream_map`, `stream_filter`



Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`
- Stream support in Python §3
  - Pre-declared functions for stream processing
    - Examples: `stream_tail`, `stream_map`, `stream_filter`
  - Refer to Specification of Python §3 (`python_3.pdf`)

stream_map and stream_filter

```
def stream_map(f, s):
    if is_none(s):
        return None
    else:
        return pair(f(head(s)),
                    lambda: stream_map(f, stream_tail(s)))

def stream_filter(p, s):
    if is_none(s):
        return None
    elif p(head(s)):
        return pair(head(s),
                    lambda: stream_filter(p,
    stream_tail(s)))
    else:
        return stream_filter(p, stream_tail(s))
```

Streams of Integers

```
def integers_from(n):
    return pair(n, lambda: integers_from(n + 1))
integers = integers_from(1)
```



Streams of Integers

```
def integers_from(n):
    return pair(n, lambda: integers_from(n + 1))
integers = integers_from(1)
```

```
def eval_stream(s, n):
    return None if n == 0 else pair(head(s),
        eval_stream(stream_tail(s), n - 1))
```

Streams of Integers

```
def integers_from(n):
    return pair(n, lambda: integers_from(n + 1))
integers = integers_from(1)
```



```
def eval_stream(s, n):
    return None if n == 0 else pair(head(s),
        eval_stream(stream_tail(s), n - 1))
```



```
eval_stream(integers, 5)
```



Streams of Integers

```
def integers_from(n):
    return pair(n, lambda: integers_from(n + 1))
integers = integers_from(1)
```



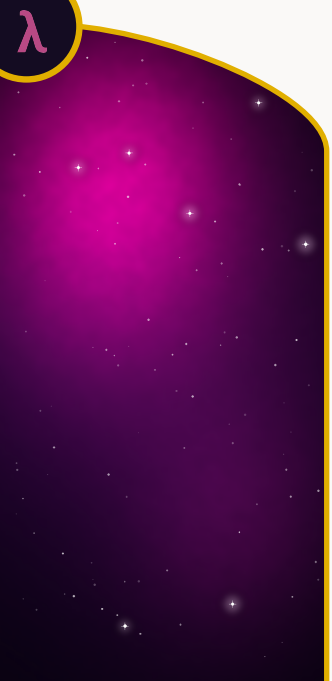
```
def eval_stream(s, n):
    return None if n == 0 else pair(head(s),
        eval_stream(stream_tail(s), n - 1))
```



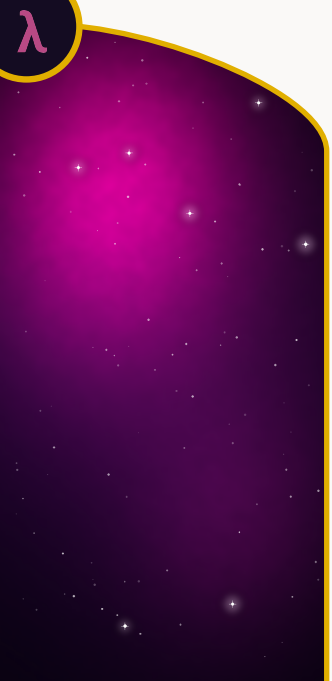
```
eval_stream(integers, 5)
```



```
[1, [2, [3, [4, [5, None]]]]]
```



Why do we need streams?



Why do we need streams?

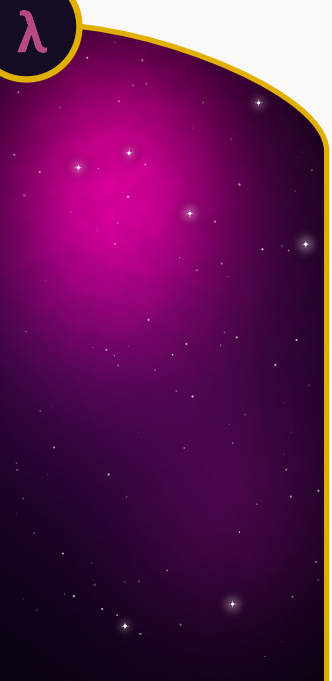
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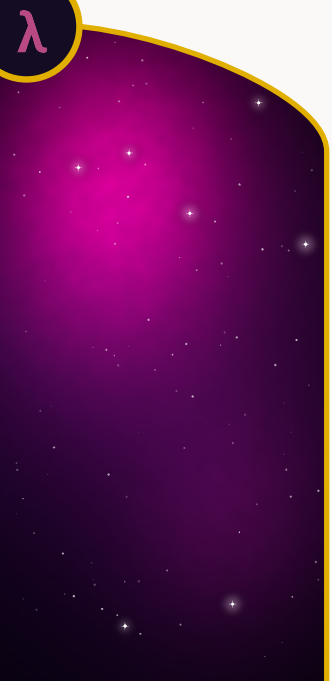


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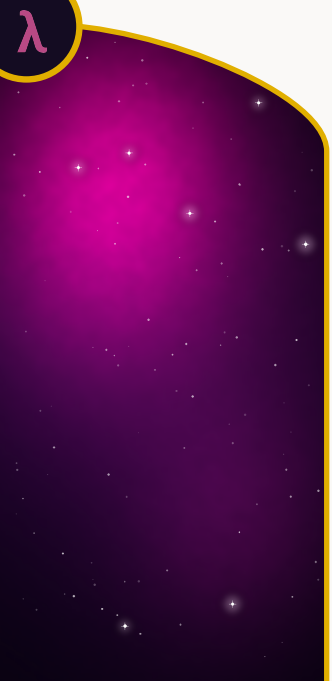
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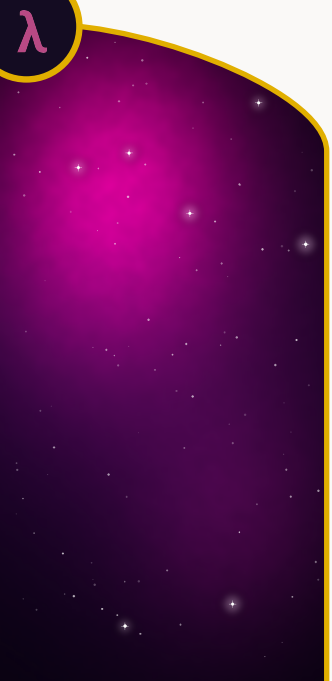
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- Example applications
- Video & Audio streaming
- Process and play A/V on-the-fly without entire file available
- Stream editing
- Process a file as a stream (e.g. find and replace text using AWK)



Outline

Review

Streams as Recursive Definitions

Memoization

Applications

Recall Stream of Ones

```
# original version
def ones_stream():
    return pair(1, ones_stream)
ones = ones_stream()
```



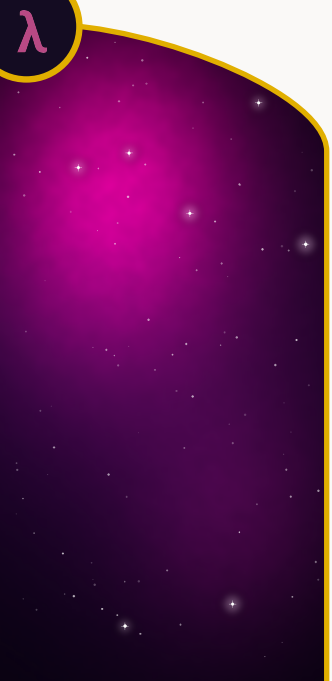
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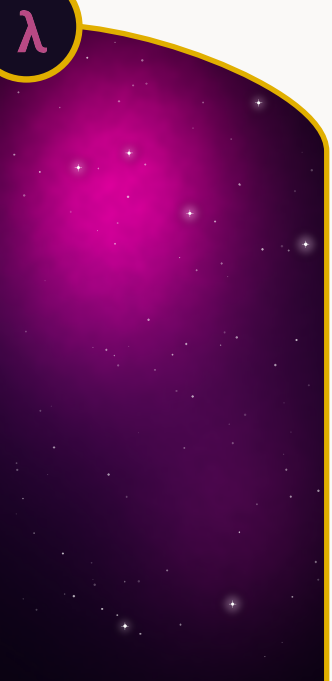


```
# alternative version
ones = pair(1, lambda: ones)
```





Defining a Repeating Sequence



Defining a Repeating Sequence

- Wanted: Stream containing 1, 2, 3, 1, 2, 3, 1, 2, 3, ...

Defining a Repeating Sequence

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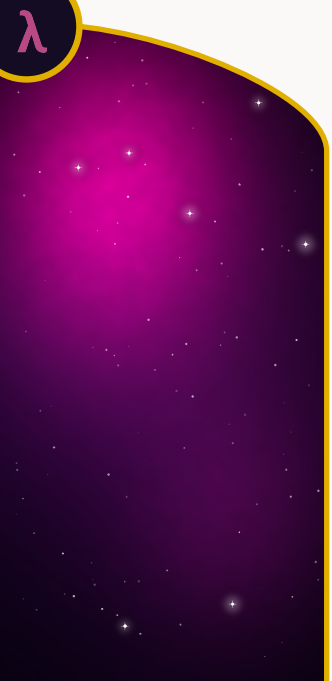
```
# A stream containing 1, 2, 3, 1, 2, 3, and so on
rep123 = pair(1, lambda: pair(2,
                        lambda: pair(3, lambda: rep123)))
eval_stream(rep123, 15)
```

Defining a Repeating Sequence

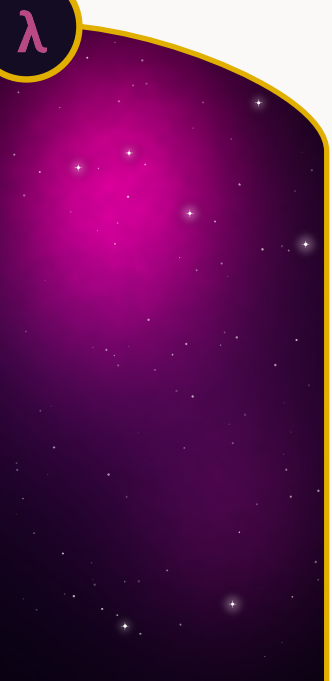
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```
# A stream containing 1, 2, 3, 1, 2, 3, and so on
rep123 = pair(1, lambda: pair(2,
                        lambda: pair(3, lambda: rep123)))
eval_stream(rep123, 15)
```

```
[1, [2, [3, [1, [2, [3, [1, [2, [3, [1, [2, [3, [1,
[2, [3, None]]]]]]]]]]]]]]
```

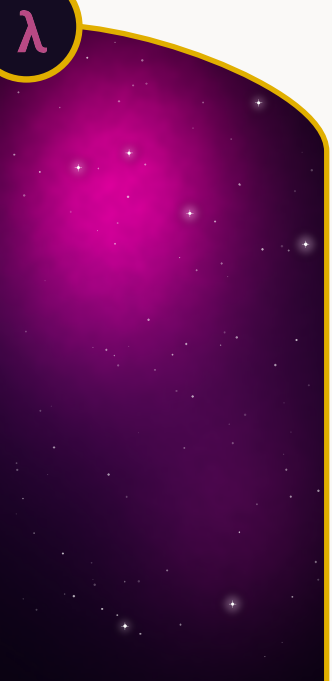


Adding Two Streams



Adding Two Streams

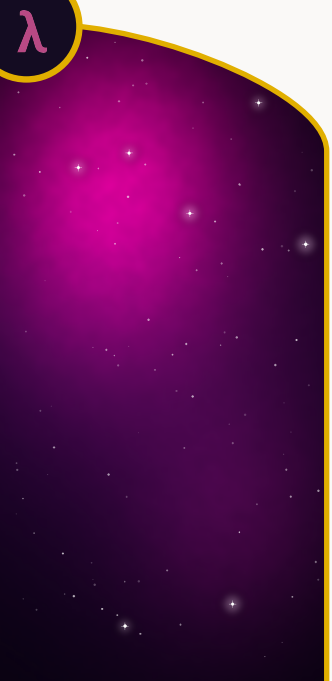
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Adding Two Streams

- Wanted: A function `add_streams` that takes two streams and returns a stream that contains the element-wise sums
- Example: Adding

```
1 2 3 4 5 6 ...
1 1 2 2 3 3 ...
```



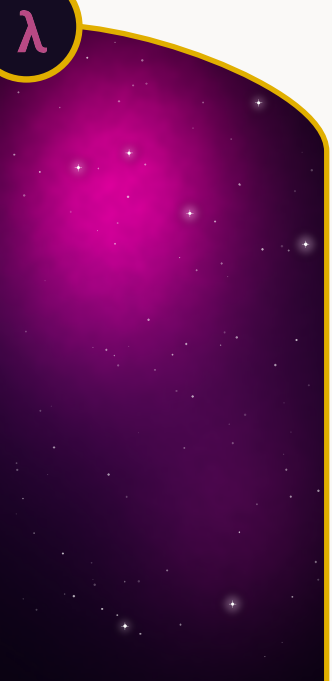
Adding Two Streams

- Wanted: A function `add_streams` that takes two streams and returns a stream that contains the element-wise sums
- Example: Adding

  ```
  1 2 3 4 5 6 ...
  1 1 2 2 3 3 ...
  ```

- should return

  ```
  2 3 5 6 8 9 ...
  ```



Adding Streams

Adding Streams

```
def add_streams(s1, s2):
    if is_none(s1):
        return s2
    elif is_none(s2):
        return s1
    else:
        return pair(head(s1) + head(s2),
                    lambda: add_streams(stream_tail(s1),
                                        stream_tail(s2)))
```



Fibonacci Numbers

```
fibs = pair(0, lambda: pair(1,
    lambda: add_streams(stream_tail(fibs), fibs)))
eval_stream(fibs, 10)
```



Fibonacci Numbers

```
fibs = pair(0, lambda: pair(1,
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eval_stream(fibs, 10)
```

```
[0, [1, [1, [2, [3, [5, [8, [13, [21, [34,
None]]]]]]]]]]
```

Fibonacci Numbers

```
        1  1  2  3  5   8  13 ... = stream_tail(fibs)
+       0  1  1  2  3   5   8 ... = fibs
-----------------------------------------------
  0  1  1  2  3  5  8  13  21 ... = fibs
```

```
fibs = pair(0, lambda: pair(1,
    lambda: add_streams(stream_tail(fibs), fibs)))
```



Integers Revisited

```
ones = pair(1, lambda: ones)
integers = pair(1, lambda: add_streams(ones, integers))
eval_stream(integers, 10)
```



Integers Revisited

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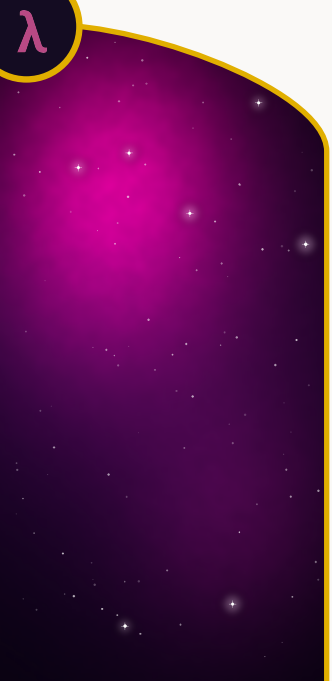
```
[1, [2, [3, [4, [5, [6, [7, [8, [9, [10,
None]]]]]]]]]]
```

Integers Revisited

```
       1  1  1  1  1  1 ... = ones
+      1  2  3  4  5  6 ... = integers
-------------------------------------
    1  2  3  4  5  6  7 ... = integers
```

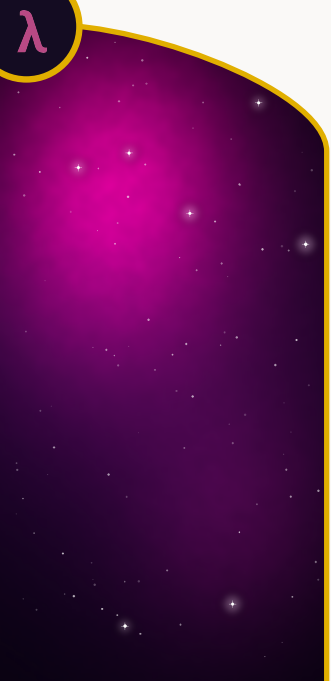
```
ones = pair(1, lambda: ones)
integers = pair(1, lambda: add_streams(ones, integers))
```





Which Stream Is This?

Which Stream Is This?



```
def mul_streams(s1, s2):
    return pair(head(s1) * head(s2),
                lambda: mul_streams(stream_tail(s1),
                                    stream_tail(s2)))
mystery = pair(1, lambda: mul_streams(mystery, integers))
eval_stream(mystery, 8)
```

Which Stream Is This?

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- Which stream does `mystery` represent?

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eval_stream(mystery, 8)
```

- Which stream does `mystery` represent?
  - The integers

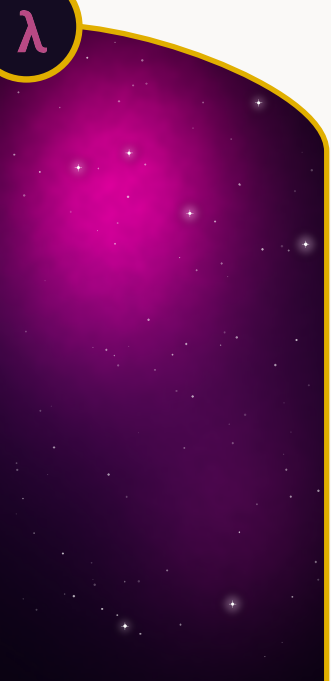
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```

- Which stream does `mystery` represent?
  - The integers
  - The Fibonacci numbers

Which Stream Is This?



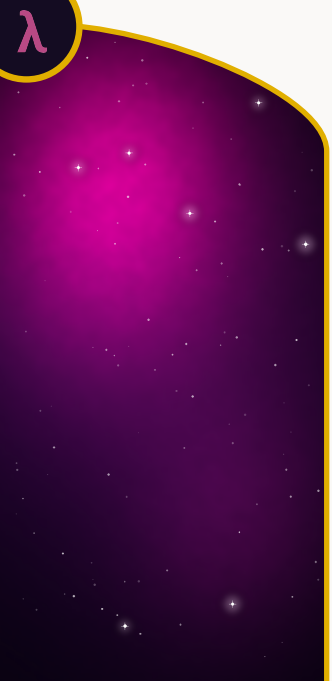
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```

- Which stream does `mystery` represent?
  - The integers
  - The Fibonacci numbers
  - The factorials

Which Stream Is This?

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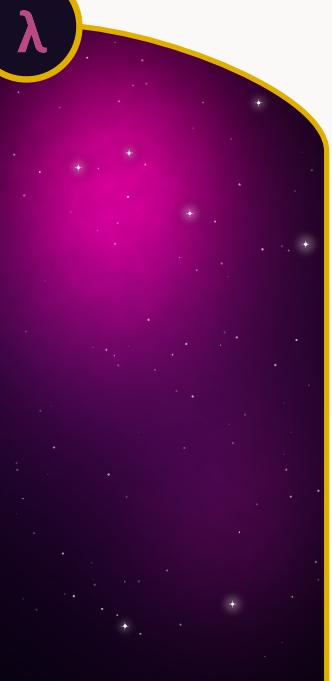
- Which stream does `mystery` represent?
  - The integers
  - The Fibonacci numbers
  - The factorials
  - The powers of 2



Check-In time!

Factor in some fun!

Look for Check-In I11A in the Source Academy!



Outline

Review

Streams as Recursive Definitions

Memoization

Applications

stream_ref

```
def enum_stream(low, hi):
    return None if low > hi else pair(low,
        lambda: enum_stream(low + 1, hi))
def stream_ref(s, n):
    return head(s) if n == 0 else stream_ref(
        stream_tail(s), n - 1)
s = enum_stream(1, 100)
```



stream_ref

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def enum_stream(low, hi):
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s = enum_stream(1, 100)
```



- stream_ref(s, 0) ⇒ 1

stream_ref

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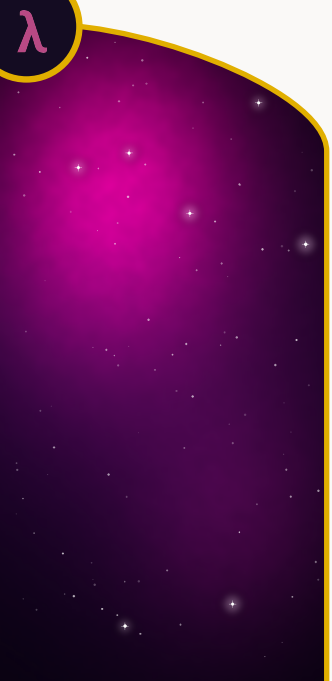
- stream_ref(s, 0) ⇒ 1
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stream_ref

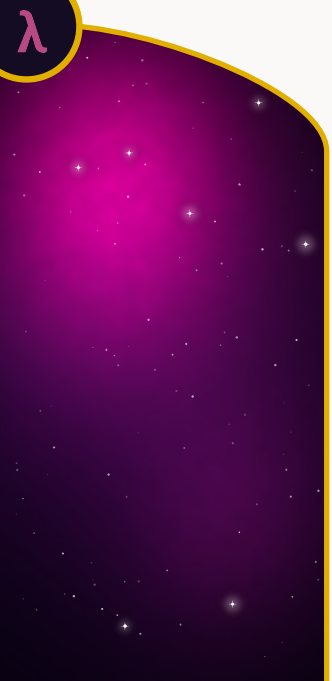
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```



- stream_ref(s, 0) ⇒ 1
- stream_ref(s, 10) ⇒ 11
- stream_ref(s, 99) ⇒ 100



Recomputing Tails



Recomputing Tails

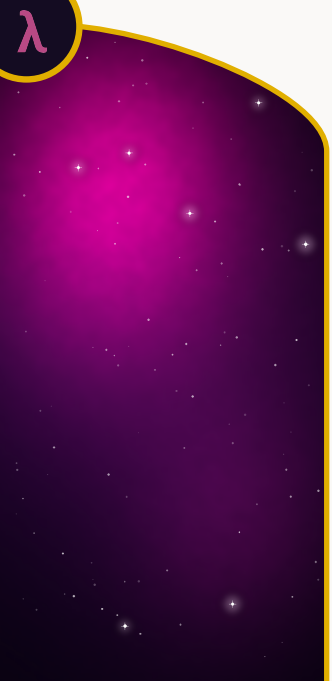
- Every time we walk past an element of a stream, its tail function runs

Recomputing Tails

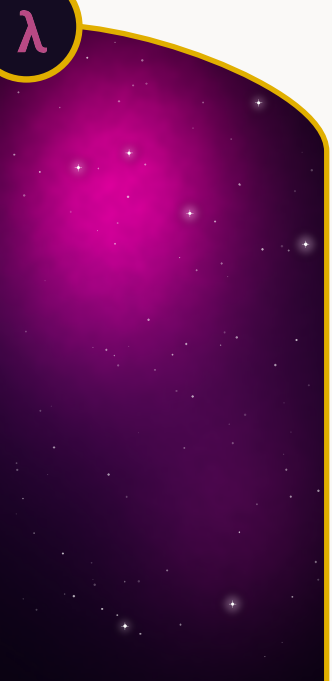
- Every time we walk past an element of a stream, its tail function runs
- Two calls of `stream_ref(s, 3)` run the same three tail functions twice

Recomputing Tails

- Every time we walk past an element of a stream, its tail function runs
- Two calls of `stream_ref(s, 3)` run the same three tail functions twice
- Idea: use *memoization* to remember the result of the first call

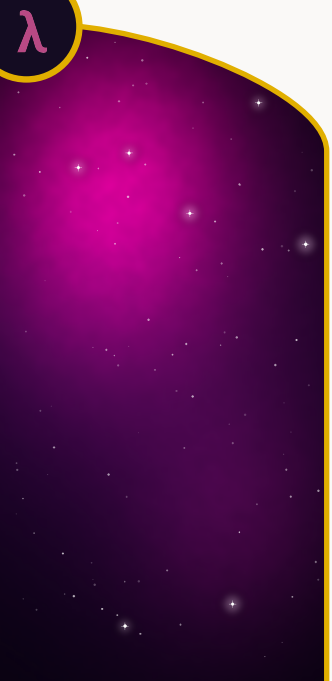


Without Memoization



Without Memoization

- `ms(m, s)` prints `m` and returns `s`



Without Memoization

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- We put it into each tail to see when the tail runs

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```
def ms(m, s):
    print(m)
    return s
def loud_from(n):
    return pair(n, lambda: ms("computing " + str(n + 1),
                              loud_from(n + 1)))
loud = loud_from(1)
stream_ref(loud, 2)
stream_ref(loud, 2)
```

Without Memoization

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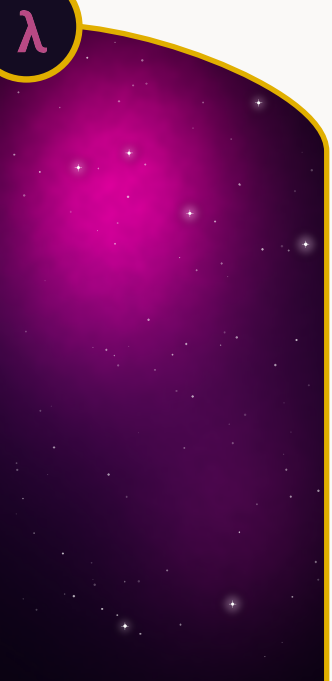
- The first call prints `computing 2` and `computing 3`

Without Memoization

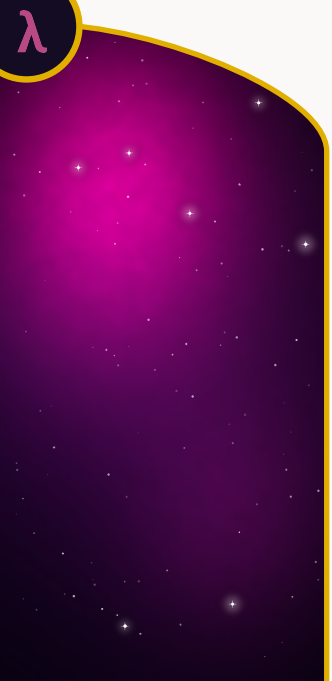
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loud = loud_from(1)
stream_ref(loud, 2)
stream_ref(loud, 2)
```

- The first call prints `computing 2` and `computing 3`
- The second call prints them again: every tail runs again

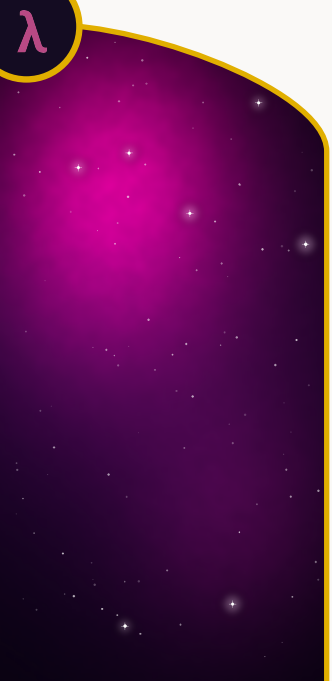


Memoization: `memo_fun`



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- `memo_fun(fun)` returns a function `mfun`



Memoization: `memo_fun`

- `memo_fun(fun)` returns a function `mfun`
  - The first call of `mfun` runs `fun` and stores the result

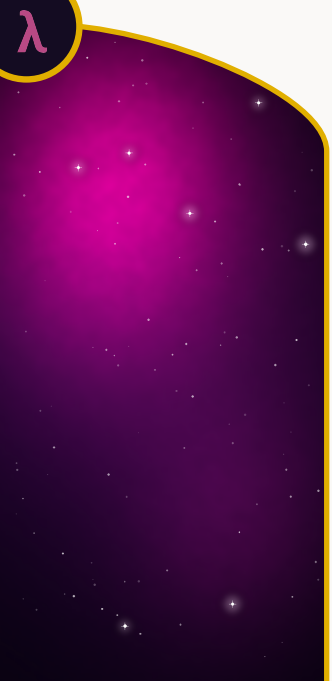
Memoization: `memo_fun`

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  - Later calls return the stored result without running `fun`

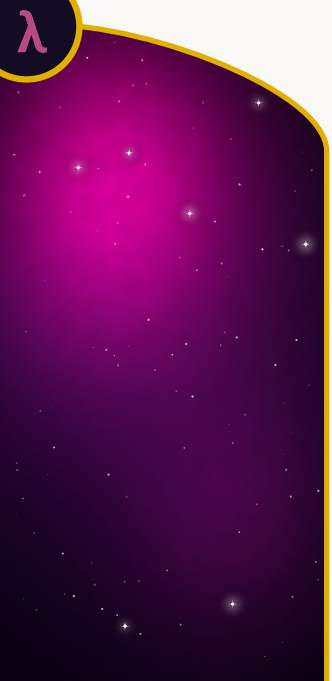
Memoization: `memo_fun`

- `memo_fun(fun)` returns a function `mfun`
  - The first call of `mfun` runs `fun` and stores the result
  - Later calls return the stored result without running `fun`

```
def memo_fun(fun):
    already_run = False
    result = None
    def mfun():
        nonlocal already_run, result
        if already_run:
            return result
        else:
            result = fun()
            already_run = True
            return result
    return mfun
```



With Memoization



With Memoization

- Each tail is wrapped in `memo_fun`

With Memoization

- Each tail is wrapped in `memo_fun`

```
def ms(m, s):
    print(m)
    return s
def memo_from(n):
    return pair(n, memo_fun(lambda: ms("computing " +
    str(n + 1),
                                        memo_from(n + 1))))
memo = memo_from(1)
stream_ref(memo, 2)
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With Memoization

- Each tail is wrapped in `memo_fun`

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def ms(m, s):
    print(m)
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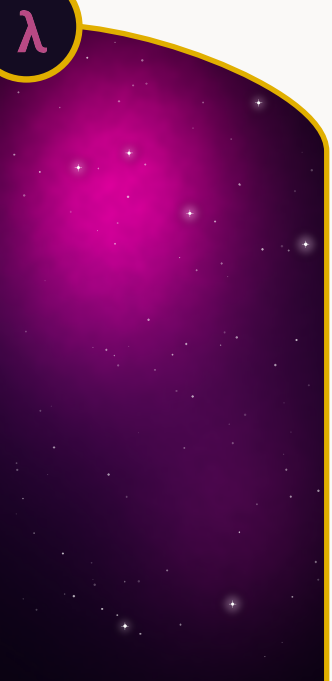
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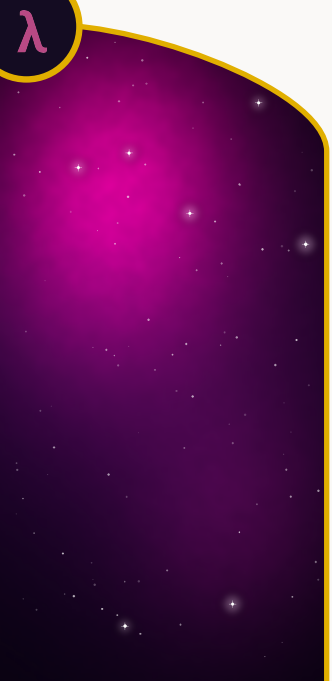
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- The first call prints `computing 2` and `computing 3`
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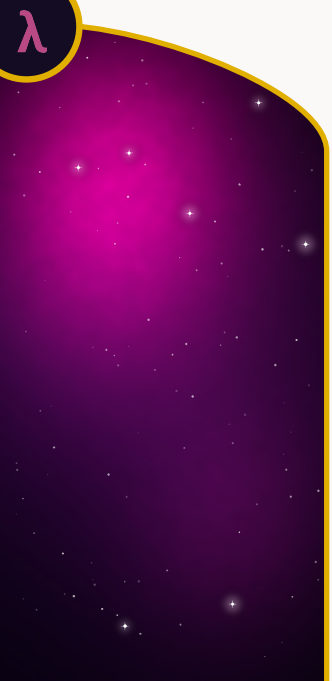


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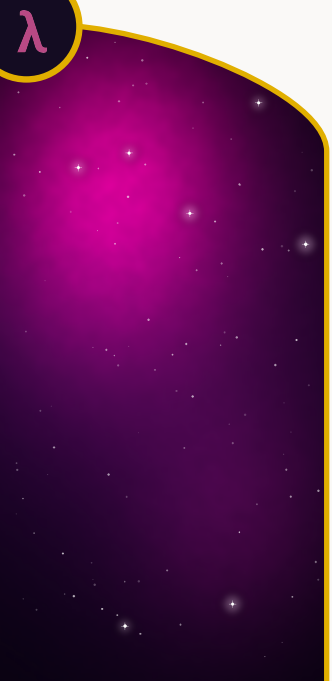
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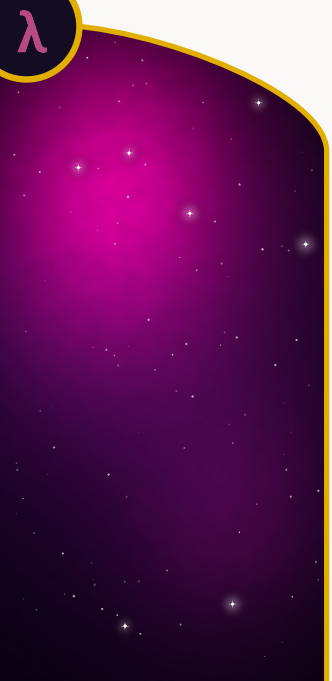
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- Pure functions have no side effects
  - They return the same value every time, so memoization never changes the result



Outline

Review

Streams as Recursive Definitions

Memoization

Applications

Example — Generating Primes

```
def is_prime(n):
    def check(d):
        return d * d > n or (n % d != 0 and check(d + 1))
    return n > 1 and check(2)
primes = pair(2, lambda: stream_filter(is_prime,
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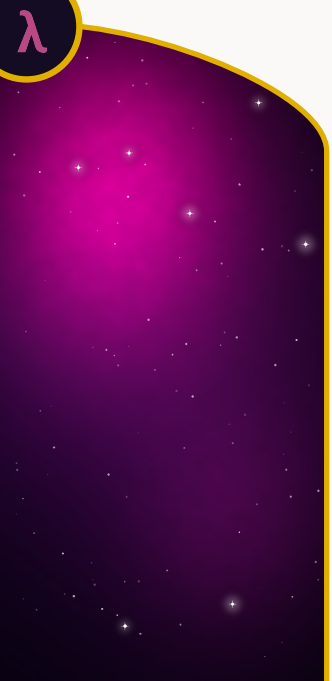
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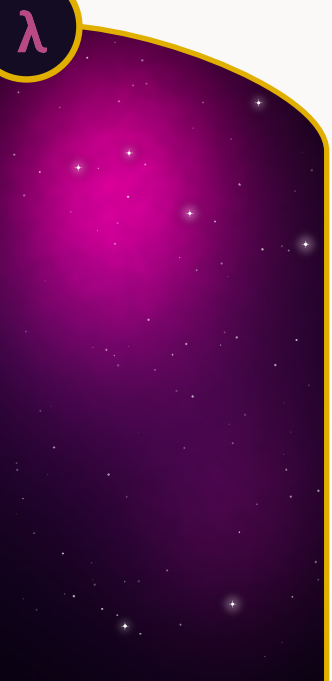
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Can we make a prime stream without using `is_prime`?



Sieving for Primes: Multiples of 2



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- Start with integers beginning with 2

  2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ...

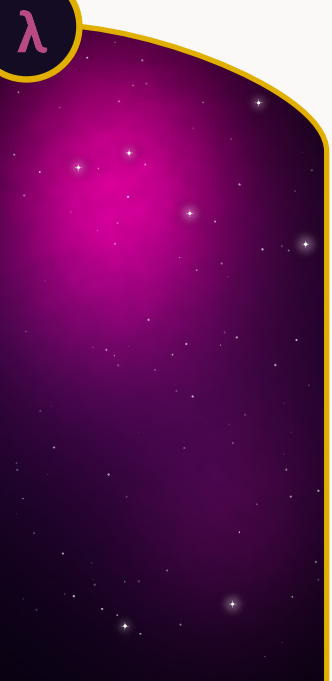
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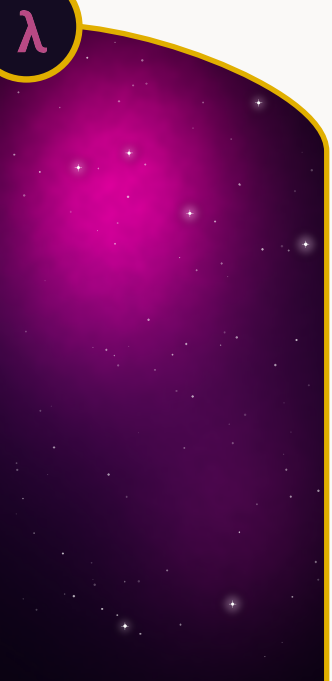
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- Filter away multiples of 2 that are greater than 2

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Sieving for Primes: Multiples of 3



Sieving for Primes: Multiples of 3

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  2 3 5 7 9 11 13 15 17 ...

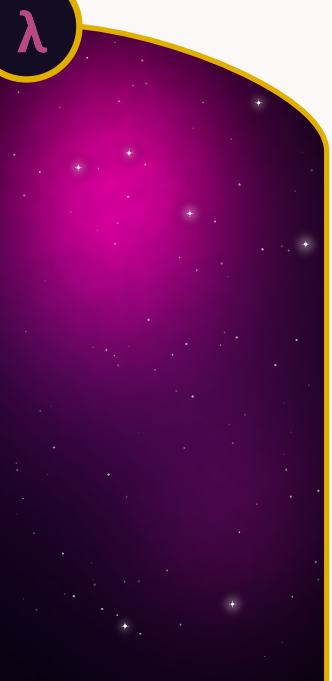
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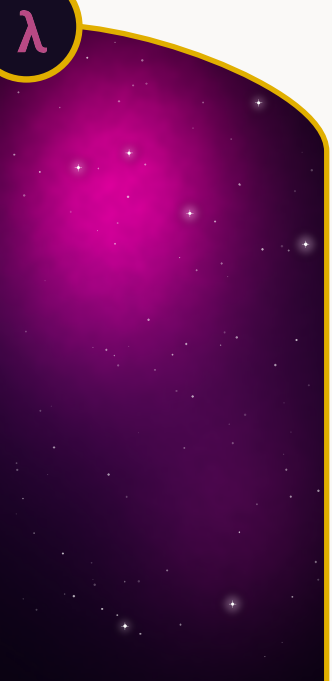
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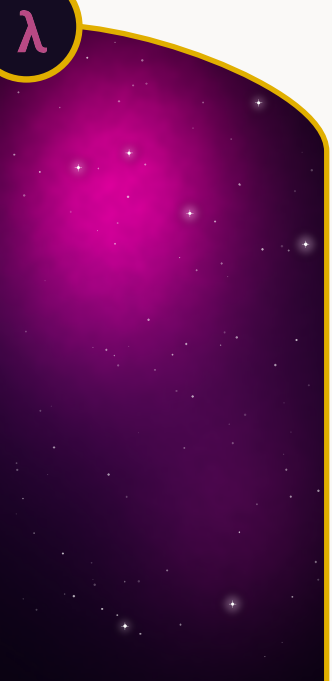
Sieving for Primes: Multiples of 5



Sieving for Primes: Multiples of 5

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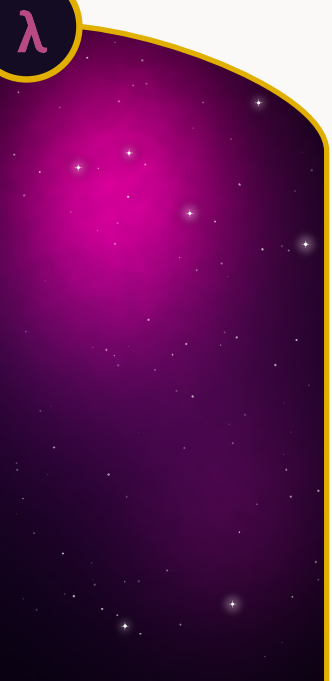


Sieving for Primes: Multiples of 5

- Next left over integer is 5

  2 3 5 7 11 13 17 ...

- And so on...



Sieving for Primes

Sieving for Primes

```
def is_divisible(x, y):
    return x % y == 0

def sieve(s):
    return pair(head(s),
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primes = sieve(integers_from(2))
eval_stream(primes, 10)
```

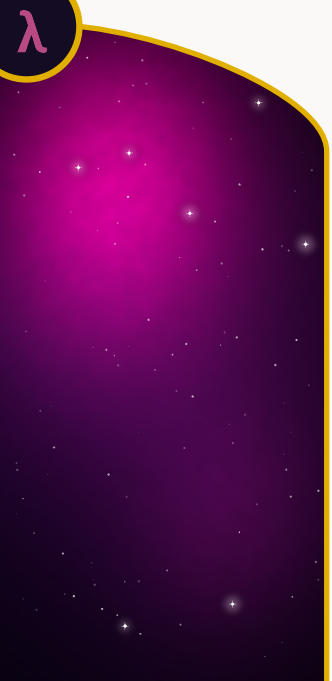
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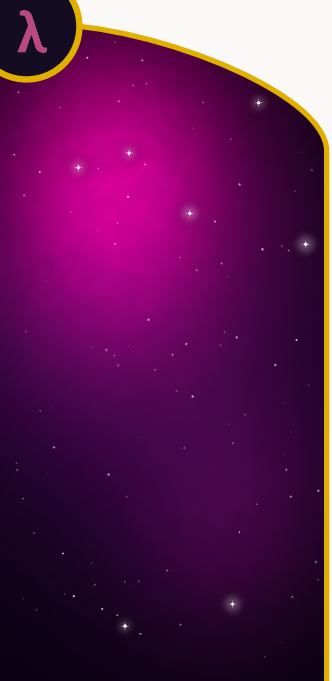
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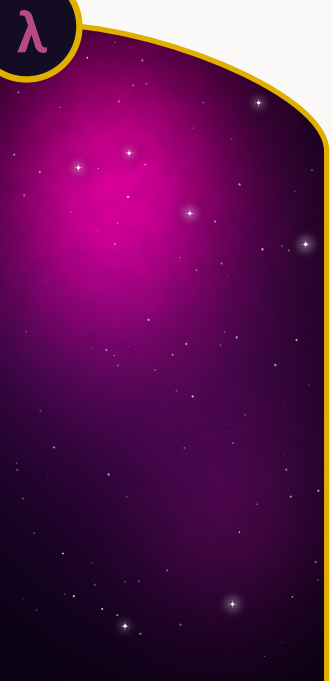


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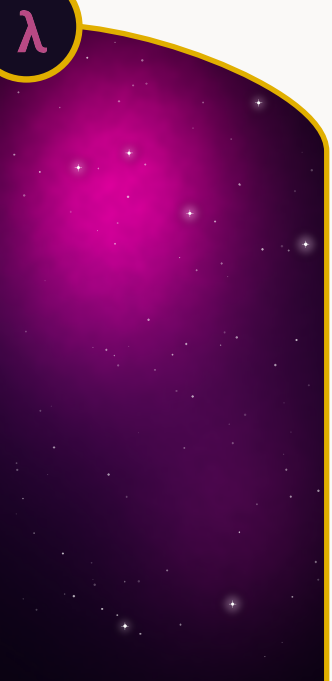


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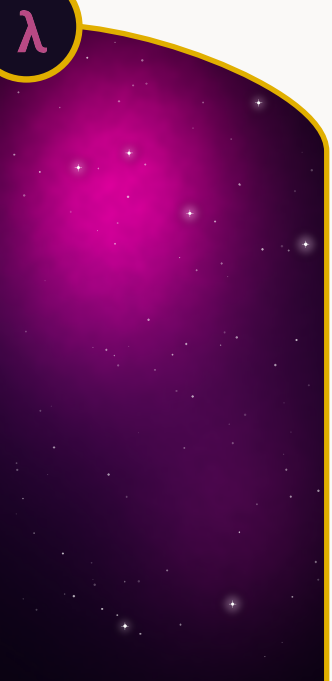
- Not actually the sieve of Eratosthenes
  - The sieve crosses off multiples of each prime
  - `sieve` tests each candidate for divisibility by the primes found so far: *trial division*, substantially less efficient (O'Neill, *The Genuine Sieve of Eratosthenes*, 2009)



Square Roots by Newton's Method

Square Roots by Newton's Method

```
def average(a, b):
    return (a + b) / 2
def good_enough(guess, x):
    return abs(guess * guess - x) < 0.001
def improve(guess, x):
    return average(guess, x / guess)
def sqrt_iter(guess, x):
    if good_enough(guess, x):
        return guess
    else:
        return sqrt_iter(improve(guess, x), x)
def sqrt(x):
    return sqrt_iter(1.0, x)
```



Using Streams for Iteration

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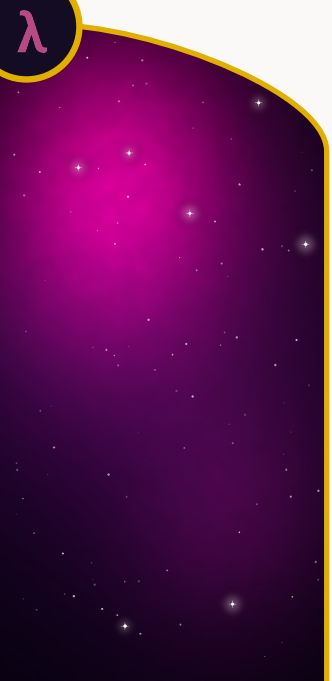
```
def sqrt_stream(x):
    guesses = pair(1.0, lambda: stream_map(
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    return guesses
eval_stream(sqrt_stream(2), 6)
```



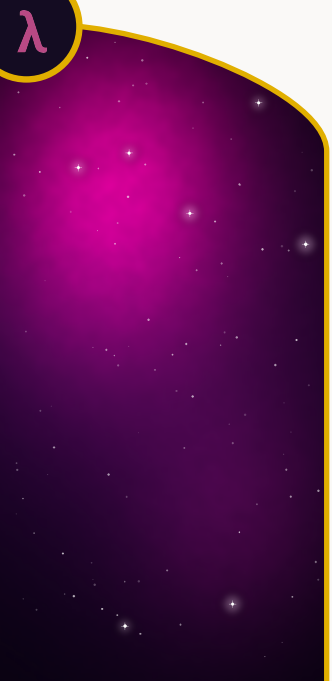
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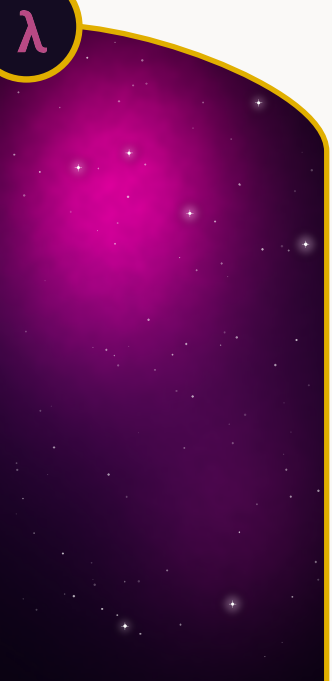


Summary



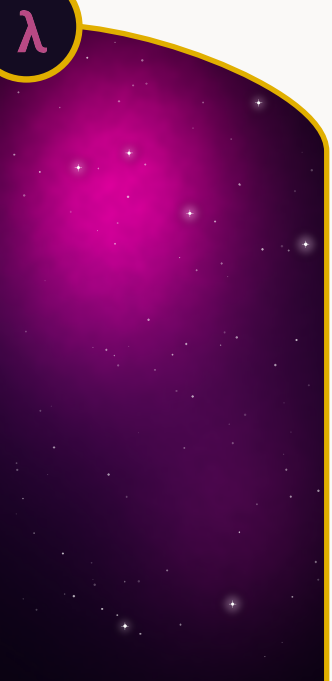
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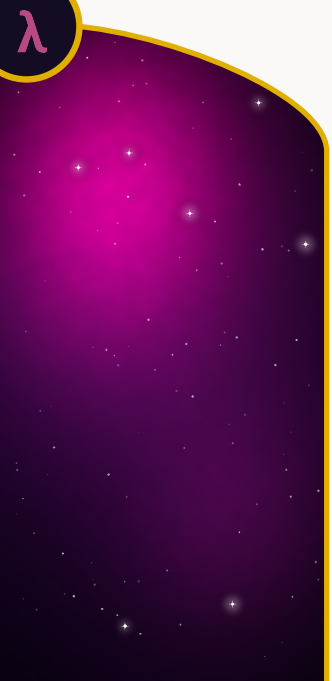
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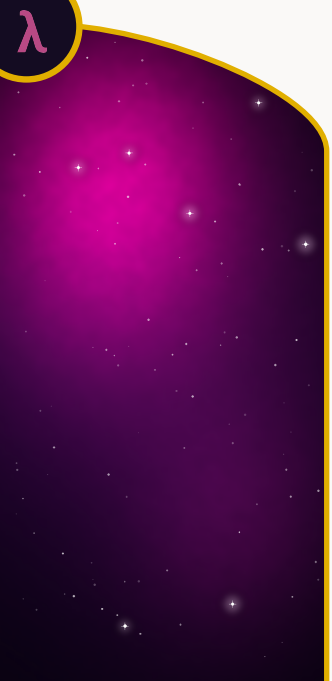
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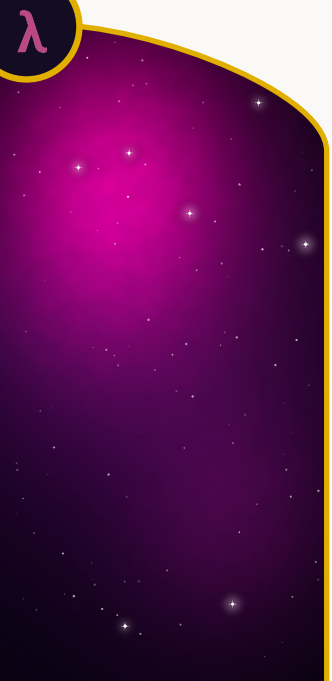
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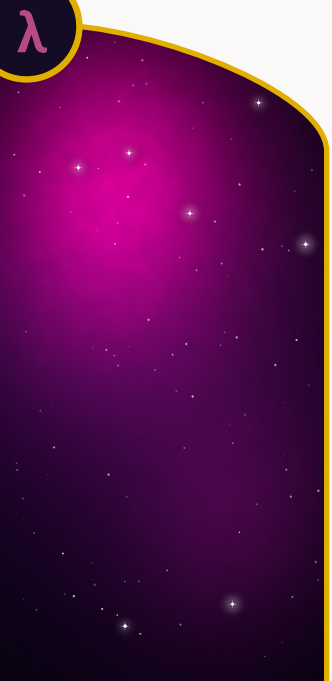
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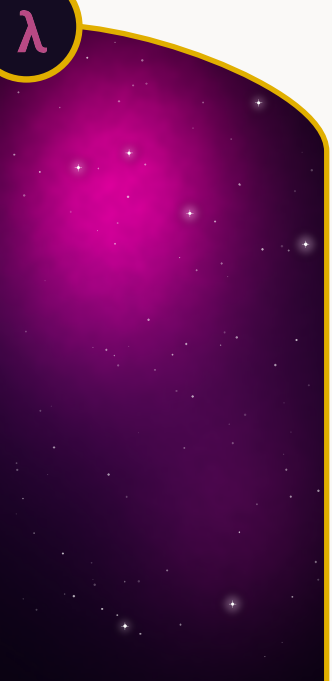
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- Streams can be used for iteration: `sieve`, `sqrt_stream`